

25PY101: Engineering Physics

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Summative preparation

CFET, QBT

Instructions

- Answer all questions.
- Show all the calculation steps clearly.
- Draw neat and properly labeled diagrams.
- Use SI units wherever necessary.

Questions

Q1. Hall effect measurement

The Hall effect measurement is shown in Figure. 1. The geometrical parameters are $d = 5 \times 10^{-3}$ cm, $W = 5 \times 10^{-2}$ cm, and $L = 0.50$ cm. The electrical parameters measured are $I_x = 0.50$ mA, $V_x = 1.25$ V, and $B_z = 650$ G. The Hall voltage is $V_H = -6.25$ mV.

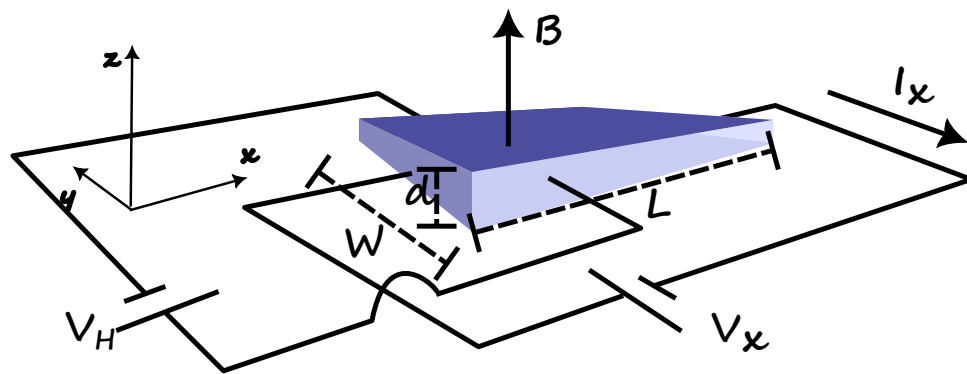


Figure 1: Hall effect measurement setup.

Determine the following material parameters.

- majority carrier type,
- carrier concentration, and
- mobility.

Q2. Graphical / Sketch-Based Question

Sketch and label the energy band diagram of

1. conductor,
2. semi-conductor, and
3. insulator.

Answer

Aim:

Topic:

Given Quantities:

Working Formulas:

Steps: